

Chapter 18

Solid Catalyzed Reactions

With many reactions, the rates are affected by materials which are neither reactants nor products. Such materials called *catalysts* can speed a reaction by a factor of a million or much more, or they may slow a reaction (negative catalyst).

There are two broad classes of catalysts: those that operate at close to ambient temperature with biochemical systems, and the man-made catalysts that operate at high temperature.

The biochemical catalysts, called *enzymes*, are found everywhere in the biochemical world and in living creatures, and without their action I doubt that life could exist at all. In addition, in our bodies hundreds of different enzymes and other catalysts are busily at work all the time, keeping us alive. We treat these catalysts in Chapter 27.

The man-made catalysts, mostly solids, usually aim to cause the high-temperature rupture or synthesis of materials. These reactions play an important role in many industrial processes, such as the production of methanol, sulfuric acid, ammonia, and various petrochemicals, polymers, paints, and plastics. It is estimated that well over 50% of all the chemical products produced today are made with the use of catalysts. These materials, their reaction rates, and the reactors that use them are the concern of this chapter and Chapters 19–22.

Consider petroleum. Since this consists of a mixture of many compounds, primarily hydrocarbons, its treatment under extreme conditions will cause a variety of changes to occur simultaneously, producing a spectrum of compounds, some desirable, others undesirable. Although a catalyst can easily speed the rate of reactions a thousandfold or a millionfold, still, when a variety of reactions are encountered, the most important characteristic of a catalyst is its *selectivity*. By this we mean that it only changes the rates of certain reactions, often a single reaction, leaving the rest unaffected. Thus, in the presence of an appropriate catalyst, products containing predominantly the materials desired can be obtained from a given feed.

The following are some general observations.

1. The selection of a catalyst to promote a reaction is not well understood; therefore, in practice extensive trial and error may be needed to produce a satisfactory catalyst.

2. Duplication of the chemical constitution of a good catalyst is no guarantee that the solid produced will have any catalytic activity. This observation suggests that it is the physical or crystalline structure which somehow imparts catalytic activity to a material. This view is strengthened by the fact that heating a catalyst above a certain critical temperature may cause it to lose its activity, often permanently. Thus present research on catalysts is strongly centered on the surface structure of solids.
3. To explain the action of catalysts, it is thought that reactant molecules are somehow changed, energized, or affected to form intermediates in the regions close to the catalyst surface. Various theories have been proposed to explain the details of this action. In one theory, the intermediate is viewed as an association of a reactant molecule with a region of the surface; in other words, the molecules are somehow attached to the surface. In another theory, molecules are thought to move down into the atmosphere close to the surface and be under the influence of surface forces. In this view the molecules are still mobile but are nevertheless modified. In still a third theory, it is thought that an active complex, a free radical, is formed at the surface of the catalyst. This free radical then moves back into the main gas stream, triggering a chain of reactions with fresh molecules before being finally destroyed. In contrast with the first two theories, which consider the reaction to occur in the vicinity of the surface, this theory views the catalyst surface simply as a generator of free radicals, with the reaction occurring in the main body of the gas.
4. In terms of the transition-state theory, the catalyst reduces the potential energy barrier over which the reactants must pass to form products. This lowering in energy barrier is shown in Fig. 18.1.
5. Though a catalyst may speed up a reaction, it never determines the equilibrium or endpoint of a reaction. This is governed by thermodynamics alone. Thus with or without a catalyst the equilibrium constant for the reaction is always the same.

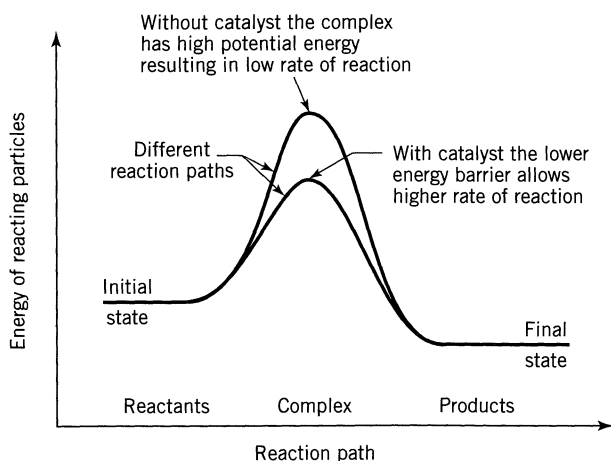


Figure 18.1 Representation of the action of a catalyst.

6. Since the solid surface is responsible for catalytic activity, a large readily accessible surface in easily handled materials is desirable. By a variety of methods, active surface areas the size of football fields can be obtained per cubic centimeter of catalyst.

Though there are many problems related to solid catalysts, we consider only those which are related to the development of kinetic rate equations needed in design. We simply assume that we have a catalyst available to promote a specific reaction. We wish to evaluate the kinetic behavior of reactants in the presence of this material and then use this information for design.

The Spectrum of Kinetic Regimes

Consider a porous catalyst particle bathed by reactant A. The rate of reaction of A for the particle as a whole may depend on:

- ① Surface kinetics, or what happens at the surfaces, interior or exterior of the particle. This may involve the adsorption of reactant A onto the surface, reaction on the surface, or desorption of product back into the gas stream.
- ② Pore diffusion resistance which may cause the interior of the particle to be starved for reactant.
- ③ Particle ΔT or temperature gradients within the particle. This is caused by large heat release or absorption during reaction.
- ④ Film ΔT between the outer surface of the particle and the main gas stream. For example, the particle may be uniform in temperature throughout but hotter than the surrounding gas.
- ⑤ Film diffusion resistance or concentration gradients across the gas film surrounding the particle.

For gas/porous catalyst systems slow reactions are influenced by ① alone, in faster reactions ② intrudes to slow the rate, then ③ and/or ④ enter the picture, ⑤ unlikely limits the overall rate. In liquid systems the order in which these effects intrude is ①, ②, ⑤, and rarely ③ and/or ④.

In different areas of application (outside of catalytic kinetics too) different combinations of these five factors enter the picture. Table 18.1 shows what we normally encounter.

Table 18.1 Factors which Influence the Rate of Reaction of Particles

Rate Influencing Factor	Porous Catalyst Particle	Catalyst Coated Surface	Burning of a Droplet of Fuel	Cells and Simple Living Creatures
① Surface reaction	Yes	Yes	No	Yes
② Pore diffusion	Yes	No	No	Maybe
③ Particle ΔT	Not too likely	No	No	No
④ Film ΔT	Sometimes	Rare	All important	No
⑤ Film mass transfer	No	Yes	All important	Could be

Although here we introduce all the phenomena which affect the rate, the real world is never so exciting that we have to concern ourselves with all five factors at any one time. In fact, in the majority of situations with porous catalyst particles we only have to consider factors ① and ②.

Here let us treat factors ① and ②; then ③ and ④ briefly.

18.1 THE RATE EQUATION FOR SURFACE KINETICS

Because of the great industrial importance of catalytic reactions, considerable effort has been spent in developing theories from which kinetic equations can rationally be developed. The most useful for our purposes supposes that the reaction takes place on an active site on the surface of the catalyst. Thus three steps are viewed to occur successively at the surface.

Step 1. A molecule is adsorbed onto the surface and is attached to an active site.

Step 2. It then reacts either with another molecule on an adjacent site (dual-site mechanism), with one coming from the main gas stream (single-site mechanism), or it simply decomposes while on the site (single-site mechanism).

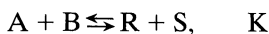
Step 3. Products are desorbed from the surface, which then frees the site.

In addition, all species of molecules, free reactants, and free products as well as site-attached reactants, intermediates, and products taking part in these three processes are assumed to be in equilibrium.

Rate expressions derived from various postulated mechanisms are all of the form

$$\text{rate of reaction} = \frac{(\text{kinetic term})(\text{driving force or displacement from equilibrium})}{(\text{resistance term})} \quad (1)$$

For example, for the reaction



occurring in the presence of inert carrier material U, the rate expression when adsorption of A controls is

$$-r_A'' = \frac{k(p_A - p_R p_S / K p_B)}{(1 + K_A p_R p_S / K p_B + K_B p_B + K_R p_R + K_S p_S + K_U p_U)^2}$$

When reaction between adjacent site-attached molecules of A and B controls, the rate expression is

$$-r_A'' = \frac{k(p_A p_B - p_R p_S / K)}{(1 + K_A p_A + K_B p_B + K_R p_R + K_S p_S + K_U p_U)^2}$$

whereas for desorption of R, controlling it becomes

$$-r_A'' = \frac{k(p_A p_B / p_S - p_R / K)}{1 + K_A p_A + K_B p_B + K K_R p_A p_B / p_S + K_S p_S + K_U p_U}$$

Each detailed mechanism of reaction with its controlling factor has its corresponding rate equation, involving anywhere from three to seven arbitrary constants, the K values. For reasons to be made clear, we do not intend to use equations such as these. Consequently, we do not go into their derivations. These are given by Hougen and Watson (1947), Corrigan (1954, 1955), Walas (1959), and elsewhere.

Now, in terms of the contact time or space time, most catalytic conversion data can be fitted adequately by relatively simple first- or n th-order rate expressions (see Prater and Lago, 1956). Since this is so, why should we concern ourselves with selecting one of a host of rather complicated rate expressions which satisfactorily fits the data?

The following discussion summarizes the arguments for and against the use of simple empirical kinetic equations.

Truth and Predictability. The strongest argument in favor of searching for the actual mechanism is that if we find one which we think represents what truly occurs, extrapolation to new and more favorable operating conditions is much more safely done. This is a powerful argument. Other arguments, such as augmenting knowledge of the mechanism of catalysis with the final goal of producing better catalysts in the future, do not concern a design engineer who has a specific catalyst at hand.

Problems of Finding the Mechanism. To prove that we have such a mechanism we must show that the family of curves representing the rate equation type of the favored mechanism fits the data *so much better than the other families that all the others can be rejected*. With the large number of parameters (three to seven) that can be chosen arbitrarily for each rate-controlling mechanism, a very extensive experimental program is required, using very precise and reproducible data, which in itself is quite a problem. We should bear in mind that it is not good enough to select the mechanism that well fits—or even best fits—the data. Difference in fit may be explainable entirely in terms of experimental error. In statistical terms these differences may not be “significant.” Unfortunately, if a number of alternative mechanisms fit the data equally well, we must recognize that the equation selected can only be considered to be one of good fit, not one that represents reality. With this admitted, there is no reason why we should not use the simplest and easiest-to-handle equation of satisfactory fit. In fact, unless there are good positive reasons for using the more complicated of two equations, we should always select the simpler of the two if both fit the data equally well. The statistical analyses and comments by Chou (1958) on the codimer example in Hougen and Watson (1947) in which 18 mechanisms were examined illustrate the difficulty in finding the correct mechanism from kinetic data, and show that even in the most carefully conducted programs of experimentation the magnitude of the experimental error will very likely mask the differences predicted by the various mechanisms.

Thus it is hardly ever possible to determine with reasonable confidence which is the correct mechanism.

Problems of Combining Resistances. Suppose that we have found the correct mechanism and resultant rate equation for the surface phenomenon. Combining this step with any of the other resistance steps, such as pore or film diffusion, becomes rather impractical. When this has to be done, it is best to replace the multiconstant rate equation by an equivalent first-order expression, which can then be combined with other reaction steps to yield an overall rate expression.

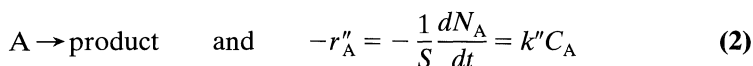
Summary on Surface Kinetics. From this discussion we conclude that it is good enough to use the simplest available correlating rate expression, hence first-order or n th-order kinetics, to represent the surface reaction.

For additional comments questioning the validity of the active-site approach, suggesting forms of kinetic equations to be used in reactor design, and suggesting what is the real utility of the active site theory, see the opposing points of view presented by Weller (1956) and Boudart (1956).

18.2 PORE DIFFUSION RESISTANCE COMBINED WITH SURFACE KINETICS

Single Cylindrical Pore, First-Order Reaction

First consider a single cylindrical pore of length L , with reactant A diffusing into the pore, and reacting on the surface by a first-order reaction



taking place at the walls of the pore, and product diffusing out of the pore, as shown in Fig. 18.2. This simple model will later be extended.

The flow of materials into and out of any section of pore is shown in detail in Fig. 18.3. At steady state a material balance for reactant A for this elementary section gives

$$\text{output} - \text{input} + \text{disappearance by reaction} = 0 \quad (4.1)$$

or with the quantities shown in Fig. 18.3,

$$-\pi r^2 \mathcal{D} \left(\frac{dC_A}{dx} \right)_{\text{out}} + \pi r^2 \mathcal{D} \left(\frac{dC_A}{dx} \right)_{\text{in}} + k''C_A(2\pi r\Delta x) = 0$$

Rearranging gives

$$\frac{\left(\frac{dC_A}{dx} \right)_{\text{out}} - \left(\frac{dC_A}{dx} \right)_{\text{in}}}{\Delta x} - \frac{2k''}{\mathcal{D}r} C_A = 0$$

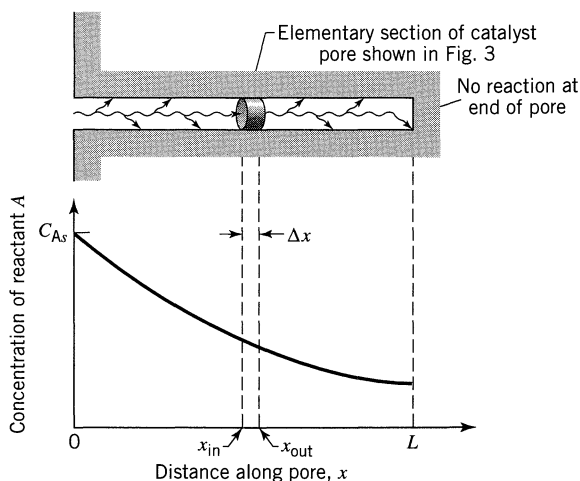


Figure 18.2 Representation of a cylindrical catalyst pore.

and taking the limit as Δx approaches zero (see the equation above Eq. 13.18a), we obtain

$$\frac{d^2 C_A}{dx^2} - \frac{2k''}{\mathcal{D}r} C_A = 0 \quad (3)$$

Note that the first-order chemical reaction is expressed in terms of unit surface area of the wall of the catalyst pore; hence k'' has unit of length per time (see

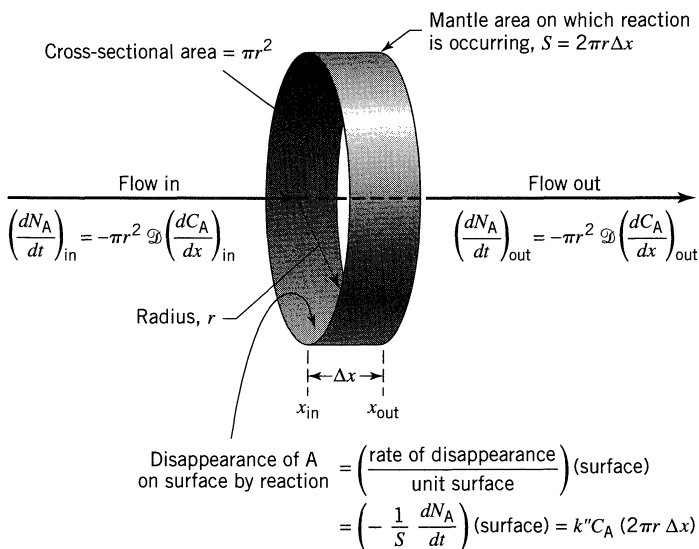


Figure 18.3 Setting up the material balance for the elementary slice of catalyst pore.

Eq. 1.4). In general, the interrelation between rate constants on different bases is given by

$$kV = k'W = k''S \quad (4)$$

Hence for the cylindrical catalyst pore

$$k = k'' \left(\frac{\text{surface}}{\text{volume}} \right) = k'' \left(\frac{2\pi r L}{\pi r^2 L} \right) = \frac{2k''}{r} \quad (5)$$

Thus in terms of volumetric units Eq. 3 becomes

$$\frac{d^2 C_A}{dx^2} - \frac{k}{\mathcal{D}} C_A = 0 \quad (6)$$

This is a frequently met linear differential equation whose general solution is

$$C_A = M_1 e^{mx} + M_2 e^{-mx} \quad (7)$$

where

$$m = \sqrt{\frac{k}{\mathcal{D}}} = \sqrt{\frac{2k''}{\mathcal{D}r}}$$

and where M_1 and M_2 are constants. It is in the evaluation of these constants that we restrict the solution to this system alone. We do this by specifying what is particular about the model selected, a procedure which requires a clear picture of what the model is supposed to represent. These specifications are called the boundary conditions of the problem. Since two constants are to be evaluated, we must find and specify two boundary conditions. Examining the physical limits of the conceptual pore, we find that the following statements can always be made. First, at the pore entrance

$$C_A = C_{As}, \quad \text{at} \quad x = 0 \quad (8a)$$

Second, because there is no flux or movement of material through the interior end of the pore

$$\frac{dC_A}{dx} = 0, \quad \text{at} \quad x = L \quad (8b)$$

With the appropriate mathematical manipulations of Eqs. 7 and 8 we then obtain

$$M_1 = \frac{C_{As} e^{-mL}}{e^{mL} + e^{-mL}}, \quad M_2 = \frac{C_{As} e^{mL}}{e^{mL} + e^{-mL}} \quad (9)$$

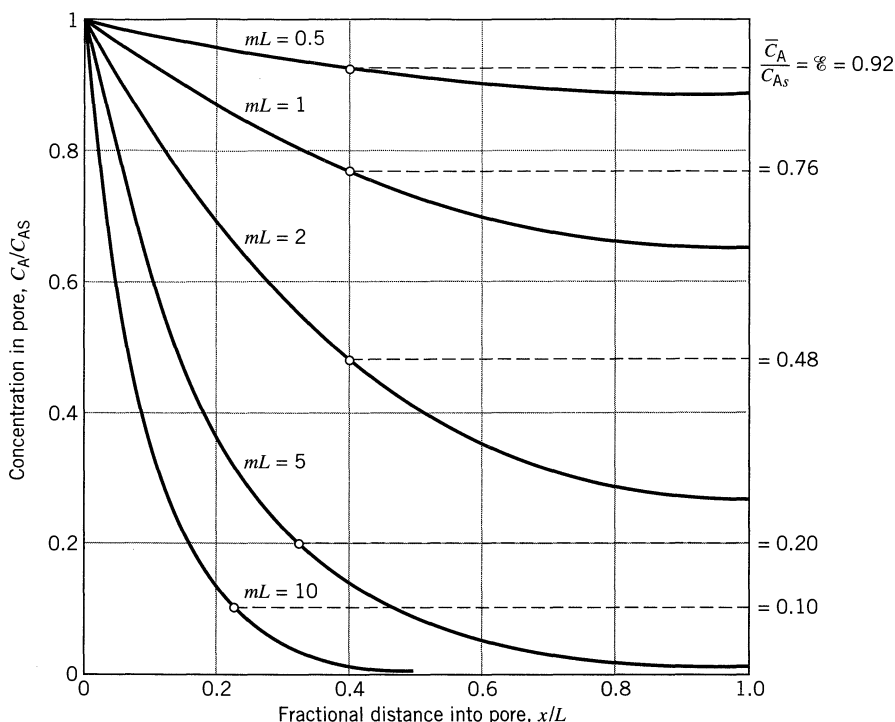


Figure 18.4 Distribution and average value of reactant concentration within a catalyst pore as a function of the parameter $mL = L\sqrt{k/\mathcal{D}}$

Hence the concentration of reactant within the pore is

$$\frac{C_A}{C_{A_s}} = \frac{e^{m(L-x)} + e^{-m(L-x)}}{e^{mL} + e^{-mL}} = \frac{\cosh m(L-x)}{\cosh mL} \quad (10)$$

This progressive drop in concentration on moving into the pore is shown in Fig. 18.4, and this is seen to be dependent on the dimensionless quantity mL , or M_T , called the *Thiele modulus*.

To measure how much the reaction rate is lowered because of the resistance to pore diffusion, define the quantity $\mathcal{E}$ called the effectiveness factor as follows:

$$\begin{aligned} \text{Effectiveness factor, } \mathcal{E} &= \frac{(\text{actual mean reaction rate within pore})}{(\text{rate if not slowed by pore diffusion})} \\ &= \frac{\bar{r}_A, \text{ with diffusion}}{r_A, \text{ without diffusion resistance}} \end{aligned} \quad (11)$$

In particular, for first-order reactions $\mathcal{E} = C_A/C_{A_s}$ because the rate is proportional to the concentration. Evaluating the average rate in the pore from Eq. 10 gives

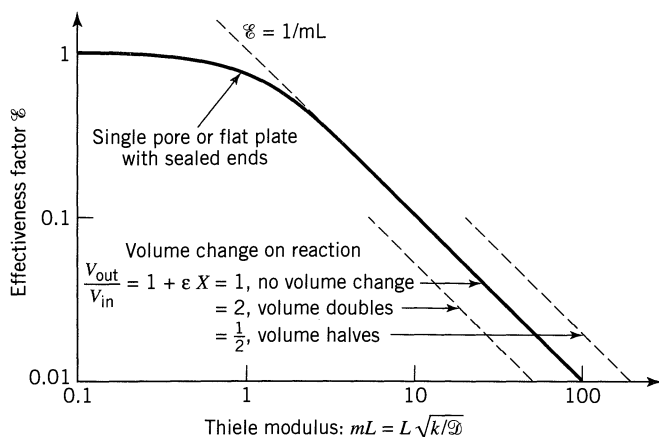


Figure 18.5 The effectiveness factor as a function of the parameter mL or M_T , called the Thiele modulus, prepared from Aris (1957) and Thiele (1939).

the relationship

$$\mathcal{E}_{\text{first order}} = \frac{\bar{C}_A}{C_{As}} = \frac{\tanh mL}{mL} \quad (12)$$

which is shown by the solid line in Fig. 5. With this figure we can tell whether pore diffusion modifies the rate of reaction, and inspection shows that this depends on whether mL is large or small.

For small mL , or $mL < 0.4$, we see that $\mathcal{E} \cong 1$, the concentration of reactant does not drop appreciably within the pore; thus pore diffusion offers negligible resistance. This can also be verified by noting that a small value for $mL = L \sqrt{k/D}$ means either a short pore, slow reaction, or rapid diffusion, all three factors tending to lower the resistance to diffusion.

For large mL , or $mL > 4$, we find that $\mathcal{E} = 1/mL$, the reactant concentration drops rapidly to zero on moving into the pore, hence diffusion strongly influences the rate of reaction. We call this the regime of *strong pore resistance*.

18.3 POROUS CATALYST PARTICLES

The results for a single pore can approximate the behavior of particles of various shapes—spheres, cylinders, etc. For these systems the following apply.

- 1. Use of the proper diffusion coefficient.** Replace the molecular diffusion coefficient $\mathcal{D}$ by the effective diffusion coefficient of fluid in the porous structure. Representative $\mathcal{D}_e$ values for gases and liquids in porous solids are given by Weisz (1959).
- 2. Proper measure of particle size.** To find the effective distance penetrated

by gas to get to all the interior surfaces we define a characteristic size of particle

$$L \left\{ \begin{array}{l} = \left(\frac{\text{volume of particle}}{\text{exterior surface available}} \right), \text{ any particle shape} \\ = \frac{\text{thickness}}{2}, \text{ for flat plates} \\ = \frac{R}{2}, \text{ for cylinders} \\ = \frac{R}{3}, \text{ for spheres} \end{array} \right. \quad (13)$$

3. Measures of reactions rates. In catalytic systems the rate of reaction can be expressed in one of many equivalent ways. For example, for first-order kinetics

$$\text{Based on volume of voids in the reactor} \quad -r_A = -\frac{1}{V} \frac{dN_A}{dt} = kC_A, \left[\frac{\text{mols reacted}}{\text{m}^3 \text{ voids} \cdot \text{s}} \right] \quad (14)$$

$$\text{Based on weight of catalyst pellets} \quad -r'_A = -\frac{1}{W} \frac{dN_A}{dt} = k'C_A, \left[\frac{\text{mol reacted}}{\text{kg cat} \cdot \text{s}} \right] \quad (15)$$

$$\text{Based on catalyst surface} \quad -r''_A = -\frac{1}{S} \frac{dN_A}{dt} = k''C_A, \left[\frac{\text{mol reacted}}{\text{m}^2 \text{ cat. surf.} \cdot \text{s}} \right] \quad (16)$$

$$\text{Based on volume of catalyst pellets} \quad -r'''_A = -\frac{1}{V_p} \frac{dN_A}{dt} = k'''C_A, \left[\frac{\text{mol reacted}}{\text{m}^3 \text{ solid} \cdot \text{s}} \right] \quad (17)$$

$$\text{Based on total reactor volume} \quad -r'''_A = -\frac{1}{V_r} \frac{dN_A}{dt} = k'''C_A, \left[\frac{\text{mol reacted}}{\text{m}^3 \text{ reactor} \cdot \text{s}} \right] \quad (18)$$

Use whichever definition is convenient. However, for porous catalyst particles rates based on unit mass and on unit volume of particles, r' and r''' are the useful measures. Hence for n th order reactions

$$\begin{aligned} -r'_A \left[\frac{\text{mol A}}{(\text{kg cat}) \cdot \text{s}} \right] &= k' C_A^n \quad \text{where} \quad k' = \left[\frac{(\text{m}^3 \text{ gas})^n}{(\text{mol A})^{n-1} (\text{kg cat}) \cdot \text{s}} \right] \\ -r'''_A \left[\frac{\text{mol A}}{(\text{m}^3 \text{ cat}) \cdot \text{s}} \right] &= k''' C_A^n \quad \text{where} \quad k''' = \left[\frac{(\text{m}^3 \text{ gas})^n}{(\text{mol A})^{n-1} (\text{m}^3 \text{ cat}) \cdot \text{s}} \right] \end{aligned} \quad (19)$$

4. In a manner similar to what was done for a single cylindrical pore, Thiele

(1939) and Aris (1957) related $\mathcal{E}$ with M_T for the various particle shapes as follows:

$$\begin{array}{lcl}
 \left. \begin{array}{l} A \rightarrow R \\ -r_A'' = k''' C_A \mathcal{E} \\ \text{mol/m}^3 \text{ cat} \cdot \text{s} \end{array} \right\} \dots \text{where } \mathcal{E} & \left\{ \begin{array}{l} = \frac{1}{M_T} \cdot \tanh M_T \\ = \frac{1}{M_T} \cdot \frac{I_1(2M_T)}{I_0(2M_T)} \quad \text{Bessel function} \\ = \frac{1}{M_T} \cdot \left(\frac{1}{\tanh 3M_T} - \frac{1}{3M_T} \right) \end{array} \right. & \begin{array}{l} \text{Flat plate} \\ \text{Cylinder} \\ \text{Sphere} \end{array}
 \end{array} \quad \begin{array}{l} (20) \\ (21) \\ (22) \end{array}$$

where

$$M_T = L \sqrt{k''' / \mathcal{D}_e} \quad (23)$$

These relationships are shown in Fig. 18.6. If you know $\mathcal{D}_e$, k''' , and L you can find the reaction rate from M_T and Fig. 18.6. However, what if you want to evaluate k''' from an experiment in which you measure a rate which could have been slowed by diffusional resistance, but which you are unsure of?

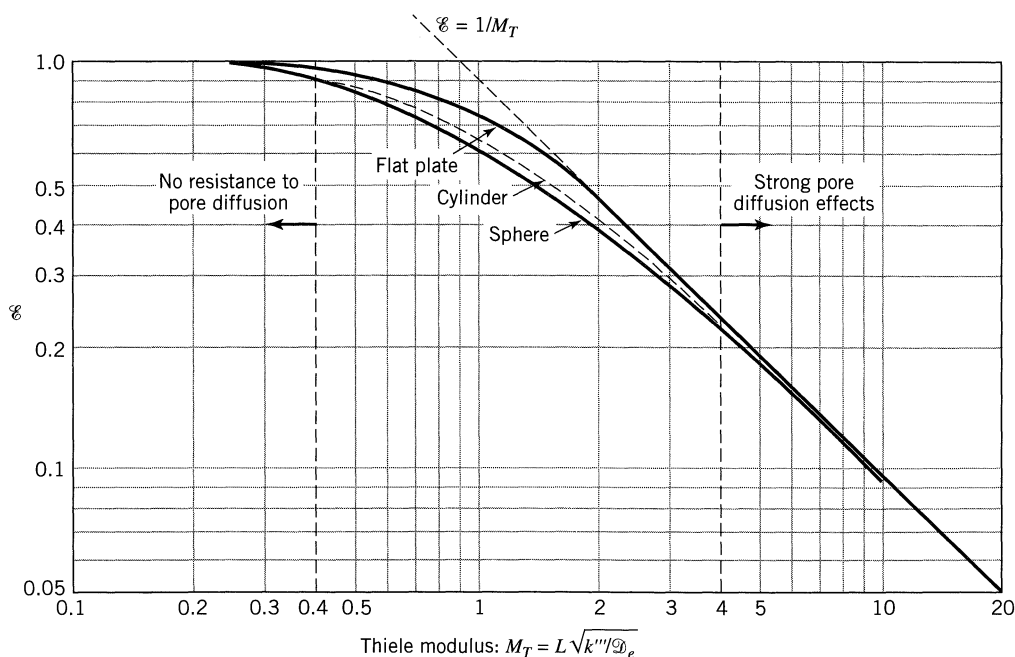


Figure 18.6 Effectiveness factor versus M_T for porous particles of various shapes.

- 5. Finding pore resistance effects from experiment.** Here we have a simple trick to help us. Define another modulus which only includes observable and measurable quantities. This is known as the Wagner-Weisz-Wheeler modulus M_W (lucky for us that the three researchers who first dealt with this problem had last names all starting with the same letter).

$$M_W = M_T^2 \mathcal{E} = L^2 \frac{(-r_A'''/C_A)_{\text{obs}}}{\mathcal{D}_e} \quad (24)$$

We shall call this the Wagner modulus.

- 6. Pore resistance limits.** When reactant fully penetrates the particle and bathes all its surfaces, then the particle is in the diffusion free regime. This occurs when $M_T < 0.4$ or $M_W < 0.15$.

At the other extreme when the center of the particle is starved for reactant and is unused then the particle is in the strong pore resistance regime. This occurs when $M_T > 4$ or $M_W > 4$.

Figures 18.6 and 18.7 show these limits.

- 7. Particles of Different Sizes.** Comparing the behavior of particles of size R_1 and R_2 we find in the diffusion-free regime

$$\frac{r'_{A1}}{r'_{A2}} = \frac{\mathcal{E}_1 k' C_A}{\mathcal{E}_2 k' C_A} = \frac{\mathcal{E}_1}{\mathcal{E}_2} = 1$$

In the regime of strong diffusion resistance

$$\frac{r'_{A1}}{r'_{A2}} = \frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{M_{T2}}{M_{T1}} = \frac{R_2}{R_1} \quad (25)$$

Thus the rate varies inversely proportional to the particle size.

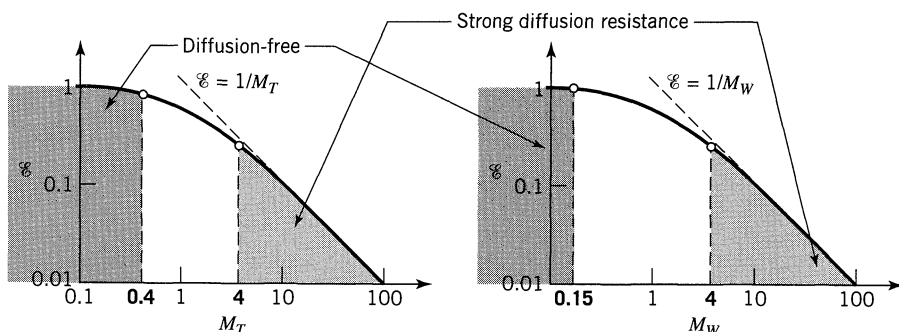


Figure 18.7 Shows the limits for negligible and for strong pore diffusion resistance.

Extensions

There are many extensions to this basic treatment. Here we just mention a few.

Mixture of Particles of Various Shapes and Sizes. For a catalyst bed consisting of a mixture of particles of various shapes and sizes Aris (1957) showed that the correct mean effectiveness factor is

$$\bar{\mathcal{E}} = \mathcal{E}_1 f'_1 + \mathcal{E}_2 f'_2 + \dots$$

where $f'_1, f'_2, \dots$ are the volume fractions of particles of sizes 1, 2, $\dots$ in the mixture.

Molar Volume Change. With decrease in fluid density (expansion) during reaction the increased outflow of molecules from the pores makes it harder for reactants to diffuse into the pore, hence lowering $\mathcal{E}$. On the other hand, volumetric contraction results in a net molar flow into the pore, hence increasing $\mathcal{E}$. For a first-order reaction Thiele (1939) found that this flow simply shifted the $\mathcal{E}$ versus M_T curve as shown in Fig. 18.5.

Arbitrary Reaction Kinetics. If the Thiele modulus is generalized as follows [see Froment and Bischoff (1962)]

$$M_T = \frac{(-r'''_{As})L}{\left[2\mathcal{D}_e \int_{C_{Ae}}^{C_{As}} (-r'''_A) dC_A\right]^{1/2}}, \quad C_{Ae} = \left(\begin{array}{c} \text{equilibrium} \\ \text{concentration} \end{array}\right) \quad (26)$$

then the $\mathcal{E}$ versus M_T curves for all forms of rate equation closely follow the curve for the first-order reaction. This generalized modulus becomes

for first-order reversible reactions

$$M_T = L \sqrt{\frac{k'''}{\mathcal{D}_e X_{Ae}}} \quad (27)$$

for n th-order irreversible reactions

$$M_T = L \sqrt{\frac{(n+1)k'''C_{As}^{n-1}}{2\mathcal{D}_e}} \quad (28)$$

n th-order reactions behave in an unexpected way in the region of strong pore resistance. Combining the n th-order rate with the generalized modulus of Eq.

28 we obtain

$$\begin{aligned} -r_A''' &= k''' C_{As}^n \mathcal{C} = k''' C_{As}^n \cdot \frac{1}{M_T} = k''' C_{As}^n \cdot \frac{1}{L} \sqrt{\frac{2\mathcal{D}_e}{(n+1)k''' C_{As}^{n-1}}} \\ &= \left(\frac{2}{n+1} \cdot \frac{k''' \mathcal{D}_e}{L^2} \right)^{1/2} C_{As}^{(n+1)/2} \end{aligned} \quad (29)$$

Thus, in the regime of strong pore diffusion, an n th-order reaction behaves like a reaction of order $(n+1)/2$ or

0 order becomes $1/2$ order

1st order remains 1st order

2nd order becomes 1.5 order

3rd order becomes 2nd order

(30)

In addition the temperature dependency of reactions is affected by strong pore resistance. From Eq. 29 the observed rate constant for n th-order reactions is

$$k_{\text{obs}}''' = \left(\frac{2}{n+1} \cdot \frac{k''' \mathcal{D}_e}{L^2} \right)^{1/2}$$

Taking logarithms and differentiating with respect to temperature and noting that both the reaction rate and to a lesser extent the diffusional process are temperature-dependent gives

$$\frac{d(\ln k_{\text{obs}}''')}{dT} = \frac{1}{2} \left[\frac{d(\ln k''')}{dT} + \frac{d(\ln \mathcal{D}_e)}{dT} \right] \quad (31)$$

With Arrhenius temperature dependencies for both reaction and diffusion we have

$$k''' = k_0''' e^{-E_{\text{true}}/RT} \quad \text{and} \quad \mathcal{D}_e = \mathcal{D}_{e0} e^{-E_{\text{diff}}/RT}$$

and replacing in Eq. 31 gives

$$\mathbf{E}_{\text{obs}} = \frac{\mathbf{E}_{\text{true}} + \mathbf{E}_{\text{diff}}}{2} \quad (32)$$

Since the activation energy for gas-phase reactions is normally rather high, say $80 \sim 240$ kJ, while that for diffusion is small (about 5 kJ at room temperature or 15 kJ at 1000°C), we can write approximately

$$\mathbf{E}_{\text{obs}} \cong \frac{\mathbf{E}_{\text{true}}}{2} \quad (33)$$

These results show that the observed activation energy for reactions influenced by strong pore resistance is approximately one-half the true activation energy.

Summary—Diffusion Resistance in Pores

For first-order surface reaction we summarize our finding in compact form in Eq. 34.

rate without any diffusion effects $= 1$ for no diffusion resistance

this flat plate expression is a good approximation for all particle shapes

... with ... $\mathcal{E} = \frac{\tanh M_T}{M_T}$

characteristic length $M_T = L \sqrt{\frac{k''' }{\mathcal{D}_e}}$

effective diffusion coefficient in porous solids; $\text{m}^3 \text{ gas/m solid-s}$

... and also ... $M_W = M_T^2 \mathcal{E} = \frac{L^2(-r'''_{A,\text{obs}})}{C_{A,\text{obs}} \mathcal{D}_e}$

where $\mathcal{E}$ = effectiveness factor, a fudge factor which varies between 0 and 1, and which accounts for the resistance to pore diffusion

M_T = Thiele modulus, useful for predicting reactor behavior from known kinetic information, thus known k'''

M_W = Weisz modulus, useful for interpreting experiments since it only includes observables

(34)

To find how pore resistance influences the rate evaluate M_T or M_W , then find $\mathcal{E}$ from the above equations or figures, and insert $\mathcal{E}$ into the rate equation.

Desirable processing range: Fine solids are free of pore diffusion resistance but are difficult to use (imagine the pressure drop of a packed bed of face powder). On the other hand a bed of large particles have a small Δp but are liable to be in the regime of strong pore diffusion where much of the pellets' interior is unused.

For most effective operations what we want is to use the largest particle size which is still free of diffusional resistance or

$$M_T \cong 0.4 \quad \text{or} \quad M_W \cong 0.15 \quad (35)$$

18.4 HEAT EFFECTS DURING REACTION

When reaction is so fast that the heat released (or absorbed) in the pellet cannot be removed rapidly enough to keep the pellet close to the temperature of the fluid, then nonisothermal effects intrude. In such a situation two different kinds of temperature effects may be encountered:

Within-particle ΔT . There may be a temperature variation within the pellet.

Film ΔT . The pellet may be hotter (or colder) than the surrounding fluid.